

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [2]: dataset=pd.read_csv("Social_Network_Ads.csv")
```

```
In [3]: dataset
```

Out[3]:

	User ID	Gender	Age	EstimatedSalary	Purchased	
	0	15624510	Male	19	19000	0
	1	15810944	Male	35	20000	0
	2	15668575	Female	26	43000	0
	3	15603246	Female	27	57000	0
	4	15804002	Male	19	76000	0
	...	...	...	...	...	...
	395	15691863	Female	46	41000	1
	396	15706071	Male	51	23000	1
	397	15654296	Female	50	20000	1
	398	15755018	Male	36	33000	0
	399	15594041	Female	49	36000	1

400 rows × 5 columns

```
In [4]: dataset = dataset.drop(columns=['User ID'])
```

```
In [5]: dataset=pd.get_dummies(dataset,drop_first=True)
```

```
In [6]: dataset
```

Out[6]:

	Age	EstimatedSalary	Purchased	Gender_Male
0	19	19000	0	True
1	35	20000	0	True
2	26	43000	0	False
3	27	57000	0	False
4	19	76000	0	True
...	...	...	...	...
395	46	41000	1	False
396	51	23000	1	True
397	50	20000	1	False
398	36	33000	0	True
399	49	36000	1	False

400 rows × 4 columns

```
In [7]: x=dataset[['Age', 'EstimatedSalary', 'Gender_Male']]
y=dataset[['Purchased']]
```

```
In [9]: from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=11)
```

```
In [15]: from sklearn.linear_model import LogisticRegression
clf = LogisticRegression(random_state=0).fit(x, y)
```

C:\Users\Gayathri R\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\utils\validation.py:1300: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
y = column_or_1d(y, warn=True)
C:\Users\Gayathri R\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\linear_model_logistic.py:469: ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
<https://scikit-learn.org/stable/modules/preprocessing.html>
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(

```
In [16]: y_pred=clf.predict(x_test)
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_test,y_pred)
print(cm)
from sklearn.metrics import classification_report
clf_report=classification_report(y_test,y_pred)
print(clf_report)
```

```
[[49  4]
 [ 5 22]]

      precision    recall  f1-score   support

     0       0.91      0.92      0.92         53
     1       0.85      0.81      0.83         27

 accuracy          0.89
 macro avg          0.88      0.87      0.87         80
```

weighted avg	0.89	0.89	0.89	80
In []:				

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

In [2]: dataset=pd.read_csv("Social_Network_Ads.csv")
```

```
In [3]: dataset
```

Out[3]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

```
In [4]: dataset = dataset.drop(columns=['User ID'])
```

```
In [5]: dataset=pd.get_dummies(dataset,drop_first=True)
```

```
In [6]: dataset
```

Out [6]:

	Age	EstimatedSalary	Purchased	Gender_Male
0	19	19000	0	True
1	35	20000	0	True
2	26	43000	0	False
3	27	57000	0	False
4	19	76000	0	True
...	...	...	...	...
395	46	41000	1	False
396	51	23000	1	True
397	50	20000	1	False
398	36	33000	0	True
399	49	36000	1	False

400 rows × 4 columns

```
In [7]: x=dataset[['Age', 'EstimatedSalary', 'Gender_Male']]
y=dataset[['Purchased']]
```

```
In [8]: from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=11)
```

```
In [9]: from sklearn.neighbors import KNeighborsClassifier
neigh = KNeighborsClassifier(n_neighbors=3)
neigh.fit(x, y)
```

C:\Users\Gayathri R\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\neighbors_classification.py:238: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().

```
return self._fit(X, y)
```

Out[9]:

KNeighborsClassifier

KNeighborsClassifier(n_neighbors=3)

```
In [11]: y_pred=neigh.predict(x_test)
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_test,y_pred)
print(cm)
from sklearn.metrics import classification_report
clf_report=classification_report(y_test,y_pred)
print(clf_report)
```

```
[[49  4]
 [ 7 20]]
```

	precision	recall	f1-score	support	
	0	0.88	0.92	0.90	53
	1	0.83	0.74	0.78	27
accuracy				0.86	80
macro avg	0.85	0.83	0.84		80

weighted avg	0.86	0.86	0.86	80
In []:				

In [1]: `import numpy as np
import pandas as pd
import matplotlib.pyplot as plt`

In [2]: `dataset=pd.read_csv("Social_Network_Ads.csv")`

In [3]: `dataset`

Out [3]:

	User ID	Gender	Age	EstimatedSalary	Purchased	
	0	15624510	Male	19	19000	0
	1	15810944	Male	35	20000	0
	2	15668575	Female	26	43000	0
	3	15603246	Female	27	57000	0
	4	15804002	Male	19	76000	0
	...	...	...	...	...	...
	395	15691863	Female	46	41000	1
	396	15706071	Male	51	23000	1
	397	15654296	Female	50	20000	1
	398	15755018	Male	36	33000	0
	399	15594041	Female	49	36000	1

400 rows × 5 columns

In [4]: `dataset = dataset.drop(columns=['User ID'])`

In [5]: `dataset=pd.get_dummies(dataset,drop_first=True)`

In [6]: `dataset`

Out [6]:

	Age	EstimatedSalary	Purchased	Gender_Male
0	19	19000	0	True
1	35	20000	0	True
2	26	43000	0	False
3	27	57000	0	False
4	19	76000	0	True
...	...	...	...	...
395	46	41000	1	False
396	51	23000	1	True
397	50	20000	1	False
398	36	33000	0	True
399	49	36000	1	False

400 rows x 4 columns

In [7]: `x=dataset[['Age', 'EstimatedSalary', 'Gender_Male']]
y=dataset[['Purchased']]`

In [8]: `from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=11)`

In [11]: `from sklearn.naive_bayes import GaussianNB
gnb = GaussianNB()
y_pred = gnb.fit(x_train, y_train).predict(x_test)
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_test,y_pred)
print(cm)
from sklearn.metrics import classification_report
clf_report=classification_report(y_test,y_pred)
print(clf_report)`

```
[[49  4]
 [ 8 19]]
      precision    recall  f1-score   support

      0       0.86      0.92      0.89         53
      1       0.83      0.70      0.76         27

 accuracy          0.84          0.81          0.83         80
 macro avg          0.84          0.81          0.83         80
 weighted avg          0.85          0.85          0.85         80
```

C:\Users\Gayathri R\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\utils\validation.py:1300: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
y = column_or_1d(y, warn=True)

In [13]: `from sklearn.naive_bayes import MultinomialNB
clf = MultinomialNB()
clf.fit(x, y)
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_test,y_pred)
print(cm)
from sklearn.metrics import classification_report
clf_report=classification_report(y_test,y_pred)
print(clf_report)`

```
[[49  4]
 [ 8 19]]
      precision    recall  f1-score   support

      0       0.86      0.92      0.89         53
      1       0.83      0.70      0.76         27

 accuracy          0.84          0.81          0.85         80
 macro avg          0.84          0.81          0.83         80
 weighted avg          0.85          0.85          0.85         80
```

C:\Users\Gayathri R\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\utils\validation.py:1300: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
y = column_or_1d(y, warn=True)

In [14]: `from sklearn.naive_bayes import ComplementNB
clf = ComplementNB()
clf.fit(x, y)
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_test,y_pred)
print(cm)
from sklearn.metrics import classification_report
clf_report=classification_report(y_test,y_pred)
print(clf_report)`

```
[[49  4]
 [ 8 19]]
      precision    recall  f1-score   support

      0       0.86      0.92      0.89         53
      1       0.83      0.70      0.76         27

 accuracy          0.84          0.81          0.85         80
 macro avg          0.84          0.81          0.83         80
 weighted avg          0.85          0.85          0.85         80
```

C:\Users\Gayathri R\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\utils\validation.py:1300: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
y = column_or_1d(y, warn=True)

In [16]: `from sklearn.naive_bayes import BernoulliNB
clf = BernoulliNB()
clf.fit(x, y)
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_test,y_pred)
print(cm)
from sklearn.metrics import classification_report
clf_report=classification_report(y_test,y_pred)
print(clf_report)`

```
[[49  4]
 [ 8 19]]
      precision    recall  f1-score   support

      0       0.86      0.92      0.89         53
      1       0.83      0.70      0.76         27

 accuracy          0.84          0.81          0.85         80
 macro avg          0.84          0.81          0.83         80
 weighted avg          0.85          0.85          0.85         80
```

C:\Users\Gayathri R\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\utils\validation.py:1300: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
y = column_or_1d(y, warn=True)

In [17]: `from sklearn.naive_bayes import CategoricalNB
clf = CategoricalNB()
clf.fit(x, y)
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_test,y_pred)
print(cm)
from sklearn.metrics import classification_report
clf_report=classification_report(y_test,y_pred)
print(clf_report)`

```
[[49  4]
 [ 8 19]]
      precision    recall  f1-score   support

      0       0.86      0.92      0.89         53
      1       0.83      0.70      0.76         27

 accuracy          0.84          0.81          0.85         80
 macro avg          0.84          0.81          0.83         80
 weighted avg          0.85          0.85          0.85         80
```

```
C:\Users\Gayathri R\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\utils\validation.py:1300: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples, ), for example using ravel().
y = column_or_1d(y, warn=True)
```

In []: